When Dividends Deceive: Mispricing in Public Real Estate Markets

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Abstract

We investigate the free dividends fallacy, the tendency of investors to irrationally overvalue dividends, in equity Real Estate Investment Trusts (REITs). REITs offer an attractive setting because the propensity to pay dividends is exogenously determined by tax law, allowing the researcher to isolate behavioral preferences from rational quality signaling. Using a comprehensive panel of REIT dividend events, we find that REIT returns are elevated on dividend paying months, consistent with price pressure from dividend-seeking investors. Investors bid up REIT share prices when dividend yields are high, interest rates are low, and past market performance has been weak. These patterns subsequently reverse, and insider trading patterns are consistent with informed exploitation. The results persist within a subset of REITs that are least likely to pay dividends absent the legal requirement, supporting the interpretation that investor demand for dividends persists even when payouts do not signal quality.

Keywords: Behavioral real estate, Free dividends fallacy, Mispricing, Real estate investment trusts, Market anomalies, Informed trading.

JEL Classifications: G12, G14, R33, G41, G35.

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1 Introduction

Traditional financial theory characterizes dividends as a redistribution of existing firm value rather than the creation of new value. According to Miller and Modigliani (1961)'s irrelevance theorem, dividends do not enhance shareholder wealth in perfect capital markets because any payment made as a dividend reduces the share price equivalently. Despite this clear theoretical understanding, empirical and behavioral finance research has documented instances where investors demonstrate irrational preferences for dividends, a phenomenon termed the "free dividends fallacy" (Hartzmark and Solomon, 2013; Hartzmark and Solomon, 2019). Investors suffering from this fallacy incorrectly perceive dividends as a costless or otherwise superior form of returns compared to capital gains.

The free dividends fallacy occurs when investors treat \$1 of dividends as a superior source of return than \$1 of capital gains. Hartzmark and Solomon (2019) suggest that this phenomenon occurs predominantly because stock declines approximately equal to the dividend are not particularly salient to investors due to the high daily volatility of equities. For example, on July 11, 2024, the stock price of Avalon Bay Communities Inc. was \$202.86, and it declared a quarterly dividend of \$1.70 (annualized dividend yield of 3.35%). Due to the dividend payment, the stock price should fall approximately 0.84% (1.7/202.86) on the ex-dividend date. However, consider a scenario where, absent the dividend, the stock price would have appreciated by 1% due to some macroeconomic or firm-specific news released on the same day. The observed return on the day would be positive 0.16%, and the investor would receive a cash dividend of \$1.70 per share. This lack of salience to the investor may lead them to believe they have received money without an actual cost.

Hartzmark and Solomon (2019) and Hartzmark and Solomon (2013) document this behavioral anomaly for a sample of dividend paying firms, showing that investors systematically overvalue dividend-paying stocks. However, in studies of dividend-related behavior, one of the central empirical challenges is the endogeneity of dividend policy within the broader universe of public firms. Dividend-paying firms are not randomly selected. Rather, they tend to be larger, more profitable, more mature, and subject to greater analyst coverage and institutional ownership (Fama and French, 2001; Denis and Osobov, 2008; Grullon et al., 2002; Hoberg and Prabhala, 2009; Baker and Wurgler, 2004).

As a result, investors' revealed preference for dividend-paying firms could reflect a rational preference for firm quality, rather than an irrational preference for dividends themselves. This concern is amplified by the fact that in the general equity universe, dividend initiation, omission, or timing decisions can carry informational content (Baker & Wurgler, 2004). Consequently, empirical attempts to identify a "dividend demand premium" may be confounded by selection effects, whereby dividends serve as proxies for unobserved or difficult-to-measure dimensions of firm quality. [1]

The real estate investment trust (REIT) sector provides a particularly advantageous empirical setting to address this identification problem. REITs are legally mandated to distribute at least 90% of taxable income as dividends in order to maintain their tax-advantaged status (Boudry, 2011; Case et al., 2012; Hardin & Hill, 2008).[2] This regulatory structure ensures that all REITs pay regular dividends, and therefore the decision to pay dividends is exogenous.

We analyze this dividend preference empirically. Our analysis uses comprehensive data on U.S. equity REITs from 1980 to 2023. We begin with a simple analysis of monthly stock returns on predicted dividend months following Hartzmark and Solomon (2013). Predicted dividend months for quarterly dividend paying firms are months t+3, t+6, t+9, and t+12 from a firm's dividend payment. We use predicted dividend months rather than actual dividend paying months because restricting the sample to months where dividends actually occurred will systematically bias returns upward due to dropping out months where dividends would have occurred if not for financial distress. We observe a strong pattern that returns on predicted dividend months are significantly higher than those in non-predicted dividend months. These superior returns are not accompanied by an increase in risk as measured by the standard deviation. A portfolio strategy that goes long predicted dividend months and short all other months generates an attractive monthly return of 0.48% (5.76% annualized) and Sharpe ratio of 0.20 (0.69 annualized). These results are consistent with the findings of Hartzmark and Solomon (2013) and provide preliminary evidence that the free dividends fallacy occurs in the REIT industry.

The propensity to pay dividends for REITs is exogenous. However, REITs can and often do pay more than 90% of taxable income in dividends, leaving the dividend amount subject to managerial discretion (Boudry, 2011; Case et al., 2012; Hardin & Hill, 2008). Following Hartzmark and Solomon (2019), we measure stock returns from one day after the dividend announcement to one day before the ex-dividend date (the "dividend window"). This ensures that any information conveyed through the dividend amount is already incorporated into prices, allowing us to isolate price pressure from dividend-seeking investors.

We examine whether price pressure is stronger when the dividend yield is higher, interest rates are lower, or past market returns have been weak. If investors have an irrational preference for dividends, it follows they

[1] Dividend-paying firms have historically outperformed non-dividend-paying firms in total returns. However, Chen and Israelov (2024) show that this outperformance is fully explained by exposures to common factors, including the quality factor. Thus, dividend payers serve as a noisy proxy for characteristics that have driven historical outperformance, and investor preference for such firms need not be irrational. This further supports the value of examining the REIT sector given their propensity to pay dividends is disconnected from firm quality.

[2] "This high dividend payout requirement means a larger share of REIT investment returns come from dividends when compared with other stocks. In fact, over the long-term, about half of listed REIT total returns have come from dividends, compared to less than one-fourth for the S&P 500." (NAREIT: https://www.reit.com/investing/investment-benefits-reits/reits-and-dividend-income)

would likely also have a preference for firms with high dividend yields. For example, a popular "subreddit" with 762 thousand followers (top 1% by size on Reddit) titled r/dividends is dedicated to dividend focused investing. The subreddit is full of posts that focus narrowly on the dividend yield of a given investment opportunity. The REIT industry is often attractive to non-dedicated REIT investors due to their high dividend yields. For example, a Motley Fool article published in May, 2025 is entitled "2 Ultra-High Yield Dividend Stocks Down About 30% to Buy Now and Hold Forever" focuses on two equity REITs the article recommends one to buy.

We examine interest rates because dividend income is often viewed as a substitute for fixed income (Hartzmark & Solomon, 2019). Therefore, when interest rates are relatively low, dividend paying stocks become increasingly attractive. Largely due to their consistent dividend streams, REITs are often viewed as a substitute for fixed income even though the total return of REITs are much more consistent with the equities they are.[3]

Finally, Hartzmark and Solomon (2019) demonstrate that when total market returns have been poor, investors view stable dividends as a more reliable source of returns. The view that the dividend component of returns is safer than the capital gains component ignores that the dividend comes out of the capital gain. This mistaken belief can lead investors to disproportionately chase dividend-paying stocks during downturns, creating price pressure unrelated to fundamentals. Accordingly, we expect this behavior to be most pronounced when recent market returns have been weak.

Our results provide strong empirical support for the free dividends fallacy in REIT markets. The preference for dividend-paying REITs is significantly stronger for higher dividend yield REITs, during low interest rate environments, and when the market has performed badly. Higher dividend yields are associated with increased subsequent returns, suggesting investors overvalue these securities. When interest rates decline and alternative fixed-income investments become less attractive, dividend-seeking behavior intensifies. Similarly, following poor market performance, investors disproportionately chase dividend-paying REITs. These findings demonstrate economically meaningful price distortions consistent with an irrational preference for dividends in the REIT industry.

One could argue that the price increases we observe between dividend announcement and ex-dividend dates reflect rational responses to new fundamental information rather than behavioral biases. To address this concern, we examine whether these price run-ups are followed by return reversals, which would indicate temporary overvaluation rather than permanent repricing. If investors are irrationally bidding up REIT

[3] "The REIT Sector is often seen as a bond proxy, given the sector's relatively stable earnings and higher dividend yields" Source: Market Watch, 2024 https://www.marketwatch.com/livecoverage/fed-rate-decision-dow-flat-as-investors-await-guidance-on-cuts/card/reit-sector-enjoys-a-unanimous-rally-as-treasury-yields-fall-H7gNPTk7Ea78asrHMlCO.

prices due to the free dividends fallacy, we should observe poor subsequent performance as prices correct back to fundamental values. We test this by regressing future market-adjusted returns on the magnitude of price increases during our dividend window. Consistent with temporary overvaluation, we find statistically significant negative future returns across all time horizons from one week to six months following the ex-dividend date, suggesting that the initial price appreciation systematically reverses over time.

It is possible that better informed market participants (such as insiders) may trade based on the temporary mispricing that arises from dividend chasing investors. An insider anticipating the price distortion would buy leading up to the dividend announcement and sell between the dividend announcement date and the ex-dividend date, and sell in the period following the ex-dividend date. Consistent with this hypothesis, we find evidence that REIT insiders buy relatively more in the 30 days before dividend announcements and sell relatively more during and after the dividend window, suggesting they systematically exploit the predictable mispricing created by the free dividends fallacy.

The evidence of return reversals and insider trading behavior supports the interpretation that dividend-induced price movements reflect temporary mispricing driven by behavioral demand. To reinforce this interpretation, we conduct an additional analysis that leverages variation within the REIT sector itself. Specifically, we focus on a subset of firms that, based on their characteristics, would be least likely to pay dividends in the absence of regulatory requirements. If the observed price effects primarily reflect rational sorting on firm quality, then these patterns should be muted or absent among REITs with traits typically associated with low dividend propensity. We split the sample at the median of firm size and profitability—two of the most reliable predictors of voluntary dividend initiation in the broader equity universe (Denis & Osobov, 2008; Fama & French, 2001; Sharma, 2011). The persistence of our results within both subsamples strengthens the case that dividend-seeking behavior, rather than firm quality, is the primary driver of the observed mispricing.

By clearly demonstrating the presence and impact of the free dividends fallacy within the regulated context of REITs, our findings challenge the effectiveness of dividend-based investment strategies that are commonly employed by investors. Notably, because REIT shareholders are disproportionately institutional (Devos et al., 2013; Hartzell et al., 2006), the persistence of the fallacy in this setting indicates that even sophisticated investor bases are not immune to behavioral biases. Moreover, our empirical design addresses concerns about endogenous dividend initiation that complicate the interpretation of findings in Hartzmark and Solomon (2013) and Hartzmark and Solomon (2019), where self-selection into dividend-paying status may confound inferences. This paper contributes to the behavioral finance literature by isolating the effects of dividend salience in a setting with exogenous payout policies and offers practical implications for investors, policymakers, and financial educators. It highlights the need for improved investor education on the economic meaning of dividends and cautions against overly simplistic, dividend-focused heuristics that may systematically lead to suboptimal investment decisions.

This behavioral anomaly also has practical implications for institutional investors who frequently cite the volatility and equity market correlations of public REITs as reasons to favor private commercial real estate investments, despite higher fees and lower average returns (Riddiough, 2022; Riddiough and Li, 2023; Chacon et al., 2025).[4] Our identification of systematic dividend-driven mispricing provides empirical support for concerns about behavioral distortions in public real estate markets that are absent from private markets, where such systematic dividend chasing cannot occur due to the illiquid nature of private real estate transactions.

The remainder of this paper is organized as follows. Section 2 describes the data and summary statistics. Section 3 provides the empirical methods and results of the analysis. Section 4 concludes.

2 Data and Summary Statistics

Our study includes a sample of U.S. equity REITs from January 1980 to December 2023. REITs are identified using S&P Global. Our primary variable of interest is the stock return from one day after the dividend announcement date to one day before the ex-dividend date. Dividend yields and information about the dividend announcement and ex-dividend dates are collected from CRSP. We remove REITs that pay monthly dividends because the period between the announcement date and the ex-dividend date is too short for the analysis. Market interest rates are measured by the 1-month T-bill collected from the Federal Reserve Bank of St. Louis (FRED). Finally, firm characteristics are obtained from Compustat. The unit of observation in our study is REIT - dividend announcement dates. Our final sample includes 210 REITs and 10,940 REIT-dividend announcements. Each of the variables is formally defined in the Appendix.

Figure 1 describes the distribution of return outcomes from one day after the dividend announcement date to one day before the ex-dividend date. The distribution appears approximately normal, with a mean centered at 0.71%. The most common outcome is a return between 0% and 4%, however, extreme returns do occur. The focus of our work is on the cross-sectional relationship between these returns and variables that proxy for high dividend demand.

[Insert Figure 1 about here]

[4] MLG Capital's CEO notes "While publicly traded REITs offer exposure to real estate, they behave more like stocks than the underlying real estate itself. This is because public REITs trade on exchanges that are subject to the same sentiment-driven swings as the broader equity markets." Source: https://mlgcapital.com/blog/navigating-2025-market-volatility/.

Panel A of Table 1 displays the sample statistics of the variables used in the baseline regression analysis. The average return between the announcement date and the ex-dividend date is 0.71%, consistent with positive demand for the stock during the dividend window on average. As expected with a series of returns, there is considerable variation. The inter-quartile range is 5.63%. To be consistent with the regressions, we present the log dividend yield in the summary statistics. The average market return and monthly interest rates are 9.3% and 0.2%, respectively. Number of Years refers to the amount of time, in years, between the announcement date + 1 and the ex-dividend date - 1. The average is 0.060, or about 22 days.

[Insert Table 1 about here]

We also examine insider trading data for a set of tests. This data is obtained from Thompson Reuters Insiders Database. We follow Shen et al. (2021) and calculate two measures of insider trading. Net Insider Purchases (NIP) is defined by the following equation.

$$NIP_{i,j,t} = \frac{\sum_{j=1}^{n} \#\text{shares buy}_{i,j,t} - \sum_{j=1}^{n} \#\text{shares sell}_{i,j,t}}{\sum_{j=1}^{n} \#\text{shares buy}_{i,j,t} + \sum_{j=1}^{n} \#\text{shares sell}_{i,j,t}} \tag{1}$$

The aggregate number of shares (i) bought by insiders (j) during the window of interest (t) minus the number of shares sold by insiders during the window of interest divided by the number of shares traded by insiders during the window of interest (Shen et al., 2021; Lakonishok and Lee, 2001; Jagolinzer et al., 2014; Adebambo et al., 2015; Gangopadhyay et al., 2019). While Shen et al. (2021) perform this calculation monthly, we are interested in three specific time windows. The time between the announcement date and the ex-dividend date, and 30 days after the ex-dividend date.

The second measure, Ownership Increase (OI), is defined by the following equation.

$$OI_{i,t} = \frac{\sum_{i=1}^{n} \#\text{insider buyer}_{i,t}}{\sum_{i=1}^{n} \#\text{insider buyer}_{i,t} + \sum_{i=1}^{n} \#\text{insider seller}_{i,t}} \tag{2}$$

The number of insiders buying the security during the window of interest divided by the number of insiders buying and selling during the window of interest. (Shen et al., 2021; Cziraki, 2018; Gangopadhyay et al., 2019).

Panel B of Table 1 displays summary statistics for the insider trading dataset, reporting the mean, standard deviation, and the 25th, 50th, and 75th percentiles for the NIP and OI variables across all three periods of interest. The mean NIP is negative, indicating that, on average, more shares are sold by insiders than bought during all three periods of interest. Similarly, the mean OI is below 50%, reflecting fewer buyers than sellers during these periods. The 25th, 50th, and 75th percentiles for NIP are either -100% or 100%, while for OI, they are 0% or 100%. This indicates that, in most time windows, insider trades tend to be entirely one-sided—either exclusively buys or exclusively sells.

Next, we turn to correlation analysis of the key variables in the study. Table 2 presents the results. Returns from the dividend announcement date to the ex-dividend date are positively correlated with dividend yield and interest rates and negatively correlated with the last 12 months market return. Importantly, each of the three explanatory variables of interest are moderately correlated. Market returns and dividend yields have a negative correlation of -0.18. This is consistent with lower prices leading to higher dividend yields. Market returns and interest rates are also negatively correlated with a coefficient of -0.11. This is expected as higher interest rates lead to lower prices through the discount rate mechanism. Finally, dividend yields are positively correlated with interest rates with a coefficient of 0.37 through a similar mechanism (higher rates implies lower prices). Given these correlations, we will examine each of the explanatory variables individually and in the same regression.

[Insert Table 2 about here]

3 Empirical Methods and Results

3.1 ***Returns in Predicted Dividend Months***

Our first set of tests examines monthly REIT stock returns during predicted dividend months. The empirical design closely follows the intuition of Hartzmark and Solomon (2013), who show that equity returns are unusually high in months when a dividend is predicted based on firms' historical payout regularity. In the context of quarterly dividend paying REITs, months t+3, t+6, t+9, and t+12 following a dividend payment are classified as predicted dividend months.[5] Returns in all other months in which the firm paid a dividend within the prior year are classified as non-predicted dividend months.

This forward-looking classification is essential for avoiding look-ahead bias. Conditioning on realized dividend payment months would mechanically exclude periods in which a dividend would normally have been paid but was suspended due to financial distress. Such months are likely to be associated with negative returns, leading to upward biased estimates of dividend month performance. By requiring only information available at time t, our approach yields an implementable trading strategy and cleanly isolates return behavior associated with dividend predictability rather than dividend realization.

[Insert Figure 2 about here]

[5] For example, if a REIT pays a dividend in January, predicted dividend months occur in April (t+3), July (t+6), October (t+9), and the following January (t+12).

Figure 2 reports mean returns and standard deviations by event month relative to the most recent dividend payment. Predicted dividend months exhibit consistently higher average returns than non-predicted months, with a clear clustering across months 3, 6, 9, and 12. Mean returns in predicted dividend months range from approximately 1.2% to 1.5%, while returns in non-predicted months are materially lower. Importantly, the standard deviation of returns is similar across predicted and non-predicted months, indicating that the elevated returns are not accompanied by higher volatility or obvious changes in risk.

Table 3 summarizes portfolio level performance for predicted dividend months, non-predicted dividend months, and a long-short strategy that buys predicted dividend months and shorts non-predicted months. The predicted dividend portfolio earns a mean monthly return of 1.50 percent with a standard deviation of 5.42 percent, yielding a Sharpe ratio of 0.22. The median return of 1.50 percent closely matches the mean, suggesting that performance is not driven by a small number of extreme observations.

In contrast, non-predicted dividend months earn a mean return of 1.02 percent with a standard deviation of 4.98 percent and a Sharpe ratio of 0.14. Although returns in these months remain positive, they are economically and statistically weaker than those observed during predicted dividend months. The resulting long-short portfolio generates an average monthly return of 0.48 percent with a standard deviation of 2.37 percent and a Sharpe ratio of 0.20. The long-short return is highly statistically significant, with a t-statistic of 4.97.

To assess whether these return patterns reflect compensation for systematic risk, Table 3 also reports alphas from the Fama French four factor model (FF4). The four factors are market, size, value, and momentum calculated following Letdin et al. (2025). The long-short strategy earns an economically meaningful FF4 alpha of 0.47 percent per month, which remains strongly statistically significant with a t-statistic of 4.51. These results indicate that excess returns in predicted dividend months are not explained by standard equity risk factors.

[**Insert Table 3 about here**]

The magnitudes of these effects closely align with those documented by Hartzmark and Solomon (2013) despite important differences in sample construction and institutional setting. Hartzmark and Solomon report mean monthly returns of 1.38 percent in predicted dividend months and 1.02 percent in non-predicted months, implying a spread of 0.36 percent. Our REIT based spread of 0.48 percent is larger and exhibits comparably strong risk adjusted performance. This finding is particularly notable given that REITs are excluded from the samples studied in Hartzmark and Solomon (2013) and Hartzmark and Solomon (2019).

Taken together, these results provide clear evidence that REIT share prices are systematically elevated during predicted dividend months. The consistency of returns across predicted dividend horizons, the similarity in volatility across portfolios, and the persistence of abnormal performance after controlling for risk factors all point toward a behavioral explanation rather than a rational risk based one. While these portfolio-level results establish the existence of predictable return patterns, they do not identify the economic conditions under which dividend seeking behavior is strongest. In the next section, we turn to multivariate cross sectional tests to examine how dividend yields, interest rates, and broader market conditions interact with predicted dividend timing to shape return predictability.

3.2 *Dividend Yields, Past Market Returns, and Interest Rates*

The multivariate analysis contains four sets of tests. The first set of tests examines the relation between returns and the dividend yield, market interest rates, and past stock market returns. Each of these variables is selected because they represent times when we expect to find the free dividends fallacy to be particularly prevalent. The most direct indicator of dividend attractiveness is the dividend yield. Dividend yields for REITs are significantly higher than the broader market average. As of June 2024, the dividend yield on the FTSE NAREIT All Equity REITs was 4.13%. The dividend yield of the S&P500 at the same time was 1.27%.[6] A dividend-seeking investor screening all equities on dividend yields will likely identify and invest in REITs and particular in those with the highest dividend yields. We expect the free dividend fallacy effect documented in our baseline result to be strongest for REITs with higher dividend yields.

REITs, and real estate more broadly, are often categorized as an asset class that sits between stocks and bonds on the risk-return spectrum. REIT dividends can be viewed as substitutes for fixed income cash flow. Therefore, when interest rates are lower, income-seeking investors are likely to seek cash distributions from REITs.

Our third variable of interest is the overall market return in the past year. When investors perceive dividends to be safer than capital gains, ignoring that dividends come directly out of the price level, it follows that investors may flock to stocks that pay higher dividends when the capital gains component of investment returns has been poor. NAREIT reports that about half of total REIT returns come from dividends, compared to less than one-fourth of total returns for the broader market (Nareit, 2025). We expect that in times of poor recent overall market returns, price pressure from dividend-seeking investors is likely to be elevated.

We rely on Ordinary Least Squares (OLS) multivariate regression analysis to test the relation between

[6]Source: NAREIT Industry Financial Snapshot for June 2024: https://www.reit.com/data-research/reit-market-data/reit-industry-financial-snapshot

returns during our select period and each of the three independent variables. We regress returns of REIT i between one day after the dividend announcement date and one day before the ex-dividend date on each of the three aforementioned variables. The dividend yield is as of the end of the month prior to the dividend announcement date, the market interest rate as of the end of the month prior to the dividend announcement date, and the 12-month cumulative market return as of the end of the month prior to the dividend announcement date. Control variables include firm characteristics of REIT i as of the fiscal year prior to the dividend announcement date, and the number of days between one day after the dividend announcement date and one day before the ex-dividend date. Fixed effects are at the firm and year level.

Results from this analysis are presented in Table 4. The first six columns examine each of the independent variables of interest individually, with and without fixed effects. We observe a positive and significant relation between dividend yield and returns. The t-statistics are quite large (5.98 and 5.76 in columns (1) and (2), respectively). Using the coefficient in Column (2), a one standard deviation increase in dividend yields leads to a 54 basis point increase in returns. In examining the relation between interest rates and the returns, we observe a negative and statistically significant relation with and without fixed effects (Columns (3) and (4)). A one standard deviation increase in interest rates corresponds to a 109 basis point decrease in returns. Turning to market returns (Columns (5) and (6)), we observe a persistently negative and significant relation with returns, with and without fixed effects. A one standard deviation increase in market returns is associated with a 33 basis point decrease in returns.

[Insert Table 4 about here]

The three independent variables of interest are also correlated with one another. For example, higher interest rates are typically associated with lower stock returns. Therefore, it is instructive to include each of the variables together in the regression to observe the incremental contribution. Columns (7) and (8) include all three variables in the same regression. In the specification without fixed effects, dividend yields continue to be positive and significant, interest rates continue to be negative and significant, but market returns loses significance. However, when fixed effects are added, all three variables retain their significance in the directions consistent with our hypotheses. In Column (8), the coefficient on Dividend Yield is marginally lower and the coefficients on interest rate and market return are marginally higher. These results provide preliminary evidence that investors are susceptible to the free dividends fallacy.

3.3 *Future Returns*

Our second set of tests focuses on the future investment performance of investors that bid up the REIT during the period we examine. One could reasonably argue that the run up of stock price we observe in the previous set of tests represents a rational pricing movement due to an unobservable factor driving valuation changes in the REIT. We examine the future return performance following the ex-dividend date to mitigate the concern that this price movement is fundamental and therefore rational.

If the price movement is rational, we expect future returns to be unrelated to the price run-up. In this hypothesis, new information was priced in the period we observe, and the future returns should only price unexpected future information. If, however, the price run-up we observe is driven by overvaluation, we expect future returns to be negatively related to the level of overvaluation. We run a regression of future market-adjusted returns at the 1-month, 3 months, and 6-month intervals on the returns that occur between one day after the dividend announcement date to one day before the ex-dividend date.

[**Insert Table 5 about here**]

Table 5 presents the results of this analysis. We observe a persistently negative and statistically significant relation between the returns that occur from the dividend announcement date to the ex-date and market adjusted returns following the dividend announcement date. From one month to six months, the coefficients increase monotonically in both magnitude and statistical significance. At the six month interval, for every 1% return between the dividends announcement date and the ex-date, the coefficient implies a 14.34 basis point decrease in future market adjusted returns (Column (3)). These results provide additional support for the free dividends fallacy hypothesis.

3.4 *Insider Trading*

Given evidence consistent with persistent mispricing of a subset of REIT stocks due to the free dividends fallacy, it is natural to investigate whether any informed parties take advantage of such mispricing. If dividend-related price pressure reflects predictable demand from dividend-motivated investors, insiders may anticipate these dynamics and adjust the timing of their trades accordingly. An informed investor seeking to take the other side of this trade would buy the stock leading up to the dividend announcement, sell between the announcement and the ex-date, and sell following the ex-date. Firm insiders are a reasonable place to look for such trading activity as insiders have intimate knowledge on the fundamental value of the firm (Baesel and Stein, 1979; Biggerstaff et al., 2020).

To test whether insiders take advantage of this mispricing, we first examine univariate patterns in insider trading sorted by the magnitude of price run-ups during the dividend window. We sort firms into terciles based on returns from one day after the announcement date to one day before the ex-date, then calculate average insider trading activity within each tercile across three time periods: 30 days before the announcement date, between the announcement and ex-dates, and 30 days after the ex-date. We employ two measures of insider trading activity: Net Insider Purchases (NIP) and Ownership Increase (OI), both drawn from the Thomson Reuters Insiders Database and calculated following Shen et al. (2021).

NIP captures the net volume of shares traded by insiders, defined as the number of shares bought minus the number of shares sold, scaled by total shares traded. This measure ranges from -100% (all selling) to +100% (all buying). OI measures the proportion of insiders who are buyers rather than sellers during the window, ranging from 0% (all sellers) to 100% (all buyers). Both measures are calculated over the specific time windows of interest rather than on a monthly basis.

An important institutional feature to note is that insiders are net sellers on average over time due to stock-based compensation and portfolio diversification needs. The mean NIP in our sample is negative across all three periods (ranging from -18.93% to -36.25%), and the mean OI is below 50% (ranging from 32.50% to 41.57%), reflecting this general tendency toward net selling. Given this baseline, when we observe insiders selling less (NIP closer to zero or positive) or a higher proportion of buyers (higher OI), it signals that insiders want to own the stock relative to their typical behavior.

Table 6 presents the results of this analysis. Panel A reports NIP across return terciles, while Panel B reports OI. The patterns are consistent with our hypothesis that insiders anticipate and exploit the dividend-driven mispricing. In the 30 days before the announcement date, firms that subsequently experience the largest price run-ups (Tercile 3) show significantly less selling activity than those with the smallest run-ups (Tercile 1). The High-Low spread is 0.20 for NIP ($t = 4.83$) and 0.11 for OI ($t = 5.17$), both highly significant. This pattern suggests insiders are positioning themselves before the announcement, reducing their typical selling or even accumulating shares when they anticipate strong dividend-driven demand.

[**Insert Table 6 about here**]

The trading behavior reverses sharply during and after the dividend window. Between the announcement and ex-dates, firms in the highest return tercile show the most selling activity (NIP of -0.46 compared to -0.23 for the lowest tercile), with a High-Low spread of -0.23 ($t = -5.56$). Similarly, the proportion of insider buyers drops significantly in Tercile 3 relative to Tercile 1 (OI of 0.28 versus 0.39), yielding a High-Low spread of -0.11 ($t = -5.42$). This pattern continues in the 30 days following the ex-date, with NIP spreads of -0.17 ($t = -3.88$) and OI spreads of -0.08 ($t = -3.76$). These results indicate that insiders systematically unwind their positions after dividend announcements when prices are elevated, taking advantage of the temporary overvaluation created by dividend-seeking investors.

To formally test the relation between dividend window returns and insider trading while controlling for firm characteristics, we conduct multivariate regression analysis. We regress each insider trading measure on returns during the dividend window, measured from one day after the announcement date to one day before the ex-date. Control variables include firm size, leverage, profitability, asset growth, and the number of days in the window, all measured as of the last fiscal year end. We include firm and year fixed effects in all specifications.

The results in Table 7 provide compelling evidence that informed insiders strategically exploit the mispricing associated with the free dividends fallacy. The findings reveal a clear pattern of opportunistic trading behavior that aligns with the hypothesis that insiders anticipate and profit from predictable price movements around dividend announcements. In the 30 days before the announcement (Columns 1-2), both NIP and OI show significant positive coefficients (0.7053 and 0.3738, respectively, both significant at the 1% level).

[**Insert Table 7 about here**]

The trading pattern reverses during and after the dividend window. Between the announcement and ex-dates (Columns 3-4), the coefficients turn significantly negative (NIP: -1.1654, OI: -0.5727, both significant at the 1% level). This indicates that when the price run-up is larger, insiders engage in more selling activity relative to their baseline. The selling behavior persists and even intensifies 30 days after the ex-date (Columns 5-6), with coefficients of -1.4601 for NIP and -0.7222 for OI. The magnitude and persistence of these negative coefficients suggest insiders are systematically unwinding positions accumulated before the announcement, capturing profits from the predictable mispricing.

The systematic nature of the trading patterns, their economic magnitude, and their statistical significance all point to insiders exploiting their informational advantages to profit from predictable dividend-driven mispricing. When combined with our earlier evidence of future return reversals, these insider trading results provide additional confirmation that the price movements we document represent temporary overvaluation driven by behavioral biases rather than rational repricing of fundamental information.

3.5 *Small and Less Profitable Firms*

A fundamental challenge in identifying behavioral biases in dividend preference lies in disentangling irrational investor behavior from rational responses to firm quality signals. In the broader equity universe, dividend payments often serve as credible signals of managerial confidence, financial stability, and reduced agency conflicts (Baker & Wurgler, 2004; Miller & Modigliani, 1961). This signaling function creates an identification problem: when investors exhibit preferences for dividend-paying stocks, it remains unclear whether this reflects genuine behavioral biases or rational responses to information asymmetries about firm quality.

Our REIT setting provides a unique empirical laboratory that largely eliminates this identification challenge. However, we can further strengthen our identification by exploiting variation in firm characteristics that would traditionally predict dividend policy in the absence of regulatory mandates. If investors truly suffer from the free dividends fallacy, rather than rationally screening for firm quality, then the behavioral bias should be equally present among REITs that would be least likely to pay dividends voluntarily.

Following the propensity-to-pay literature pioneered by Fama and French (2001), the two most consistent predictors of voluntary dividend payments are firm size and profitability. In the general equity universe, larger and more profitable firms are significantly more likely to initiate and maintain dividend payments, while smaller and less profitable firms typically retain earnings for growth opportunities or lack sufficient cash flows for sustainable distributions (Denis & Osobov, 2008; Sharma, 2011). Critically, these characteristics serve as proxies for the very firm quality dimensions that dividends are thought to signal in voluntary settings.

To test whether our documented patterns reflect genuine behavioral biases rather than rational quality screening, we examine whether the free dividends fallacy persists among REITs that would be least likely to pay dividends in a voluntary regime. We construct interaction terms between our three main explanatory variables (dividend yield, interest rates, and past market returns) and indicator variables for firms below the median in size and profitability. Under the rational signaling hypothesis, we would expect dividend preferences to be concentrated among larger, more profitable REITs that resemble traditional voluntary dividend payers. Conversely, under the behavioral hypothesis, the free dividends fallacy should manifest equally among smaller, less profitable REITs, since these firms would never voluntarily signal through dividends yet still generate the same behavioral responses.

The results in Table 8 provide compelling evidence supporting the behavioral interpretation. Panel A presents our baseline regression specifications with interaction terms for both size and profitability. The main effects for dividend yield remain positive and statistically significant across all specifications, ranging from 0.0104 to 0.0132 with t-statistics consistently above 3.5. Importantly, the interaction terms between dividend yield and both size and profitability indicators are typically positive, small in magnitude, and statistically insignificant, suggesting that dividend preferences are not concentrated among firms that would traditionally be expected to pay dividends voluntarily.

[Insert Table 8 about here]

The pattern for interest rates provides additional support for the behavioral hypothesis. The main effects remain negative and significant (coefficients ranging from -1.2110 to -5.9439), indicating that dividend demand increases when alternative fixed-income investments become less attractive. The interaction terms with size and profitability are negative and insignificant, demonstrating that this pattern holds across all REIT types regardless of their traditional dividend-paying characteristics.

For past market returns, we observe an interesting pattern. The main effects are negative and significant in the specification with profitability interactions. In the specification without fixed effects (Column (1)), the interaction with profitability is positive and significant at the 5% level. This indicates that in this specification, the results are stronger for more profitable REITs. However, in the most robust model with fixed effects, the interaction becomes insignificant (Column (2), t-stat 0.79).

Turning to the interactions with size (Columns (3) and (4)), the main effects are no longer significant. However, the interaction with size shows a significant negative coefficient (-0.0292 with t-statistic of -2.78 in Column (3) and -0.0251 with t-statistic of -2.11 in Column (4)), suggesting that smaller REITs experience even stronger dividend demand following market downturns. This result is difficult to reconcile with rational quality screening, as smaller firms would typically be viewed as riskier during market stress, yet they exhibit stronger, not weaker, dividend-driven price appreciation.

To rigorously test whether the effects we document in Table 8 are present in smaller and less profitable firms, we perform a joint significance test of whether the main effect and interaction combined are equal to 0. Panel B reports joint significance tests for the combined effects of main variables and their interactions. The F-statistics confirm that dividend yield effects remain economically and statistically significant even after accounting for firm size and profitability (F-statistics ranging from 13.5 to 27.0, all significant at the 1% level). Similarly, interest rate effects maintain their significance across firm types, with F-statistics between 8.97 and 14.40. For the Market Return, three of the four specifications have significant F-statistics; the one exception is the profitability interactions without fixed effects (Column (1)). The fact that these relationships persist for small and less profitable firms provides strong evidence that investor behavior is driven by the behavioral mechanisms underlying the free dividends fallacy rather than rational responses to quality signals.

This analysis makes several important contributions to the literature. First, it addresses a key limitation in prior behavioral finance studies by demonstrating that dividend preferences persist even when dividends cannot credibly signal firm quality. Second, it validates the external validity of findings from Hartzmark and Solomon (2019) in a setting where endogenous dividend policy cannot confound the results. Finally, it provides clean evidence that the anomalous price patterns we document reflect genuine market inefficiencies rather than rational responses to information asymmetries.

4 Conclusion

This study provides compelling evidence for the existence and economic significance of the free dividends fallacy in the REIT market, demonstrating that behavioral biases can persist even in specialized market segments with sophisticated investor bases. By exploiting the unique regulatory structure of REITs, which mandates dividend distributions regardless of firm quality, we identify dividend preferences that are unlikely to be attributed to rational signaling mechanisms. Our analysis reveals that REIT returns are higher in months where they are expected to pay dividends. Additionally, REIT returns from one day after the dividend announcement to one day before the ex-dividend date are significantly positively related to dividend yields, negatively related to interest rates, and negatively related to past market performance. These patterns are consistent with investors incorrectly treating dividends as a superior form of return relative to capital gains, leading to predictable price distortions that are systematically exploited by informed insiders.

The robustness of our findings across multiple dimensions strengthens confidence in the behavioral interpretation. Future returns analysis confirms that the documented price run-ups represent temporary overvaluation, with negative and statistically significant reversals extending up to six months following the ex-dividend date. Insider trading patterns provide further validation, showing that informed parties strategically position themselves to profit from these predictable mispricing episodes. Most critically, our analysis of small and less profitable REITs, firms that would be highly unlikely to pay dividends absent regulatory mandates, demonstrates that the free dividends fallacy operates independently of any rational quality signaling function. Our results highlight opportunities for systematic alpha generation through positioning around dividend events.

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Figure 1: Distribution of Returns

This figure shows the distribution of the primary dependent variable: returns measured from one day after the dividend announcement date to one day before the ex-dividend date. The sample includes REITs that pay quarterly dividends from 1981 to 2023

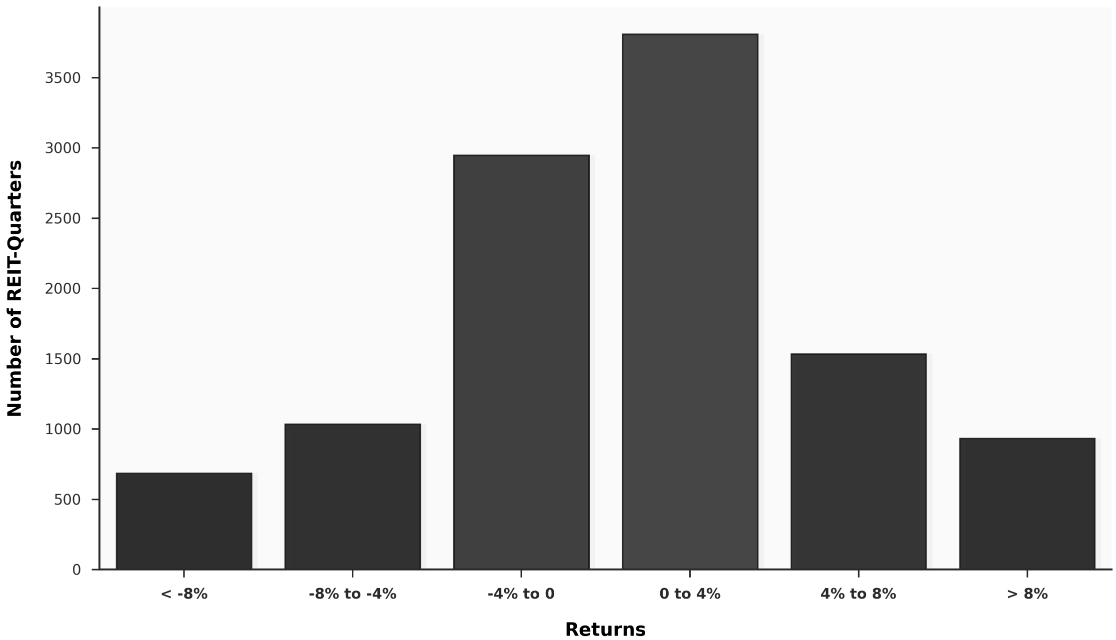

Figure 2: Returns by Month after Dividend Payment

This figure shows the average returns grouped by month following a dividend payment. Predicted dividend months are those on months 3, 6, 9 and 12 for our sample of quarterly paying dividend REITs. The left axis is returns and the right axis is the standard deviation.

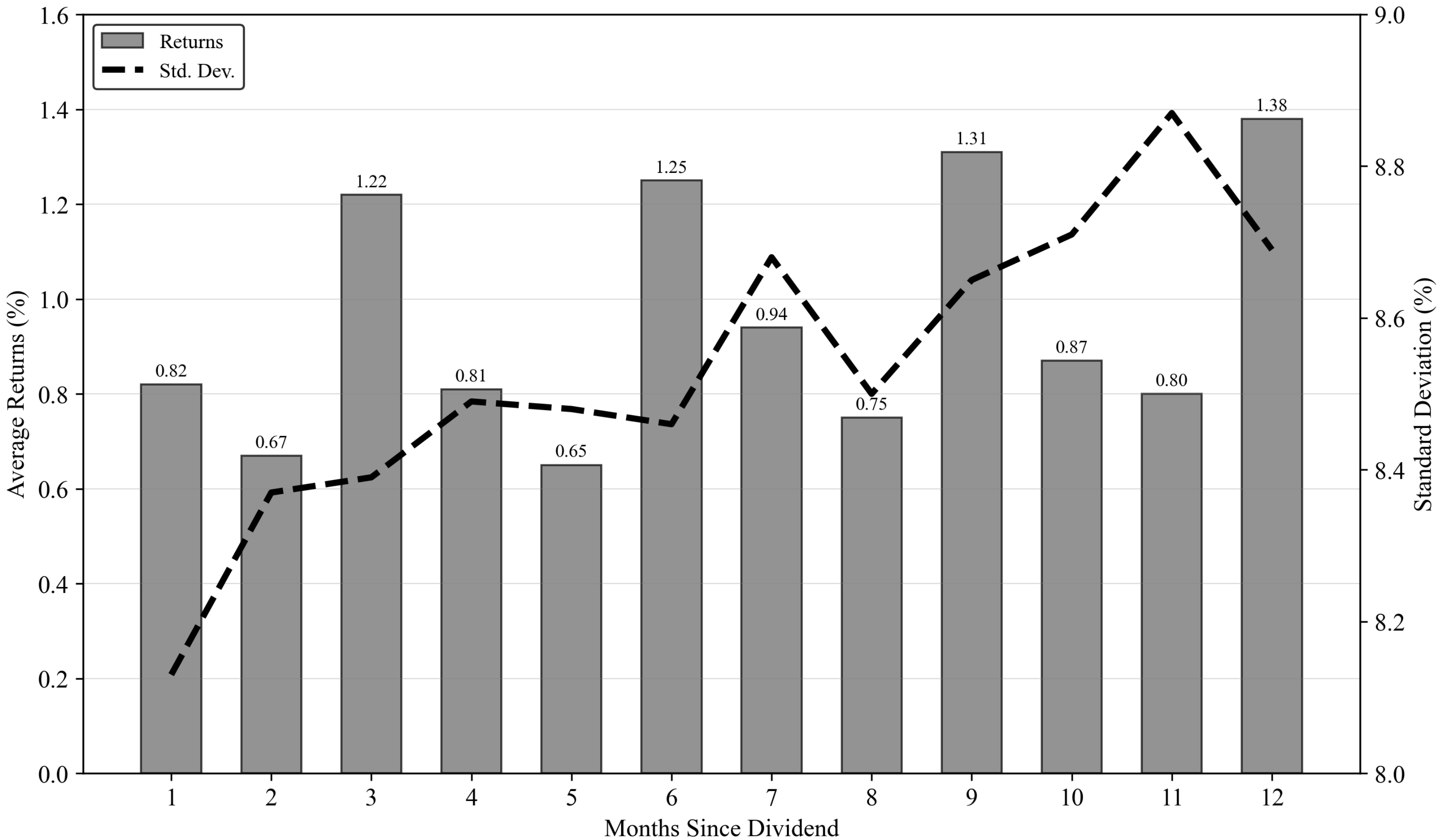

Table 1: Summary Statistics

Table 1 presents summary statistics for each variable used throughout the study. Panel A displays each of the variables in the baseline analysis. Panel B introduces insider trading variables. For insider trading variables, we measure each variable one day after the announcement date to one day before the ex-date (between AD and ExD), 30 days after the Ex-Date, and 60 days after the Ex-Date. All variables are defined in the Appendix.

Panel A: Main Regression Variables	N	Mean	Std Dev	25th Pctl	50th Pctl	75th Pctl
Returns (After Ann. to Ex-date)	10,940	0.71%	6.91%	-2.05%	0.72%	3.58%
Dividend Yield	10,940	-1.942	0.451	-2.264	-1.914	-1.603
Market Return	10,940	0.093	0.143	0.002	0.111	0.191
Interest Rate (monthly)	10,940	0.002	0.002	0.000	0.002	0.004
Log Assets	10,940	7.316	1.645	6.261	7.627	8.502
Leverage	10,940	0.522	0.164	0.437	0.528	0.626
FFO / AT	10,940	0.059	0.026	0.041	0.058	0.073
Asset Growth	10,940	0.133	0.207	-0.003	0.060	0.189
Number of Days	10,940	0.060	0.045	0.027	0.041	0.079
Panel B: Insider Trading Variables	**N**	**Mean**	**Std Dev**	**25th Pctl**	**50th Pctl**	**75th Pctl**
NIP						
30 Days before AD	2,726	-35.30%	91.85%	-100.00%	-100.00%	100.00%
Between AD and ExD	2,578	-36.25%	90.89%	-100.00%	-100.00%	100.00%
30 Days after ExD	2,729	-18.93%	96.40%	-100.00%	-100.00%	100.00%
60 Days after ExD	4,528	-26.45%	93.91%	-100.00%	-100.00%	100.00%
OI						
30 Days before AD	2,726	33.25%	45.70%	0.00%	0.00%	100.00%
Between AD and ExD	2,578	32.50%	45.17%	0.00%	0.00%	100.00%
30 Days after ExD	2,729	41.57%	47.73%	0.00%	0.00%	100.00%
60 Days after ExD	4,528	38.18%	46.47%	0.00%	0.00%	100.00%

Table 2: Pearson Correlations

This table presents Pearson correlations for each variable used throughout in the baseline analysis. Each variable is defined in the appendix.

Variable	Returns	Dividend Yield	Market Return	Interest Rate	Size	Leverage	Profitability	Asset Growth
Dividend Yield	0.08							
Market Return	−0.02	−0.18						
Interest Rate	0.02	0.37	−0.11					
Size	−0.07	−0.38	0.01	−0.53				
Leverage	0.00	−0.01	−0.02	−0.14	0.25			
Profitability	0.03	0.11	−0.03	0.13	−0.11	−0.11		
Asset Growth	0.01	−0.02	0.07	0.15	−0.06	−0.08	−0.12	
Number of Years	0.02	−0.23	0.01	−0.18	0.14	0.03	0.03	−0.04

Table 3: Dividend Returns Analysis

This table reports summary statistics for portfolios based on predicted dividend months versus other dividend-paying firms. The Long-Short portfolio goes long predicted months (months t+3, t+6, t+9, and t+12 from dividend payment in month t) and short all other months (months t+1, t+2, t+4, t+5, t+7, t+8, t+10, and t+11). The reported Mean Return and Standard Deviation use raw returns. The Sharpe ratio and Fama-French 4-Factor alpha analysis uses excess returns (raw returns minus the risk-free rate) consistent with theory. The four factors are calculated within the REIT universe and are the Market, Size, Value, and Momentum factors. T-statistics are calculated for whether the returns or alphas are different from zero.

Category	Mean return	Std. dev.	Sharpe	T-stat	Median return	FF4 alpha	T-stat FF4
Predicted dividend month (1)	1.50%	5.42%	0.22	6.26	1.50%	0.93%	5.45
All other companies with a dividend in the last 12 months (2)	1.02%	4.98%	0.14	4.64	1.24%	0.46%	3.65
Long–Short Portfolio (1) – (2)	0.48%	2.37%	0.20	4.97	0.38%	0.47%	4.51

Table 4: Dividend-Window Returns and Proxies for Dividend Demand

This table presents results from regressions of returns on dividend yield, interest rate, and past market returns. Returns are calculated from one day after the dividend announcement date to one day before the ex-dividend date (dividend window). White robust standard errors are used for each specification. Statistical significance at the 1%, 5%, and 10% confidence are identified by *,**, and ***, respectively. Each variable is defined in the appendix.

	Dependent Variable: Returns (After Ann. to Ex-date)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dividend Yield	0.0114***	0.0120***					0.0124***	0.0118***
	(5.98)	(5.76)					(6.34)	(5.66)
Interest Rate			-1.0804***	-5.4419***			-1.7593***	-6.1781***
			(-2.64)	(-2.65)			(-4.14)	(-2.97)
Market Return					-0.0099*	-0.0231***	-0.0060	-0.0239***
					(-1.95)	(-2.73)	(-1.17)	(-2.78)
Size	-0.0022***	-0.0012	-0.0040***	-0.0017**	-0.0034***	-0.0017**	-0.0031***	-0.0012
	(-4.84)	(-1.56)	(-7.82)	(-2.28)	(-8.07)	(-2.27)	(-5.91)	(-1.59)
Leverage	0.0053	0.0060	0.0084**	0.0083*	0.0080*	0.0084*	0.0051	0.0061
	(1.28)	(1.26)	(2.01)	(1.74)	(1.93)	(1.76)	(1.23)	(1.29)
Profitability	0.0447	0.0156	0.0671**	0.0198	0.0582**	0.0193	0.0548**	0.0168
	(1.61)	(0.51)	(2.43)	(0.64)	(2.10)	(0.62)	(1.98)	(0.55)
Asset Growth	0.0050*	0.0038	0.0054*	0.0022	0.0046	0.0023	0.0075**	0.0038
	(1.69)	(1.10)	(1.80)	(0.65)	(1.54)	(0.65)	(2.48)	(1.10)
Number of Years	0.0756***	0.0899***	0.0495**	0.0759***	0.0546**	0.0761***	0.0698***	0.0888***
	(3.11)	(3.36)	(2.05)	(2.84)	(2.30)	(2.85)	(2.83)	(3.31)
Observations	10,940	10,940	10,940	10,940	10,940	10,940	10,940	10,940
Adjusted R-squared	0.011	0.044	0.008	0.041	0.007	0.041	0.013	0.046
Firm FE	No	Yes	No	Yes	No	Yes	No	Yes
Year FE	No	Yes	No	Yes	No	Yes	No	Yes

Table 5: Returns following the Dividend Window

This table presents results from regressions of future market-adjusted returns on the cumulative returns from one day after the dividend announcement date to one day before the ex-dividend date (Dividend Window). Each column represents a different time period for future returns ranging from 1 month to 6 months. Each specification uses firm and year fixed effects. White robust standard errors are used. Statistical significance at the 1%, 5%, and 10% confidence are identified by *,**, and ***, respectively. Each variable is defined in the appendix.

	Dependent Variable: Market Adjusted Returns		
	1 Month	3 Months	6 Months
Returns	-0.0335**	-0.0742***	-0.1434***
	(-2.35)	(-3.20)	(-4.31)
Size	-0.0005	-0.0031**	-0.0055***
	(-0.63)	(-2.43)	(-2.85)
Leverage	-0.0034	0.0089	0.0237**
	(-0.65)	(1.06)	(2.04)
Profitability	-0.0622*	-0.0471	-0.0437
	(-1.85)	(-0.83)	(-0.52)
Asset Growth	-0.0121***	-0.0155**	-0.0137
	(-3.30)	(-2.48)	(-1.59)
Number of Years	0.0106	0.0396	0.0309
	(0.70)	(1.57)	(0.85)
Observations	9,985	9,807	9,465
Adjusted R-Sq.	0.009	0.027	0.054
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Table 6: Dividend Returns Analysis

This table presents portfolio results based on terciles of firms' returns from one day after the dividend announcement date to one day before the ex-dividend date (divided window). For each return tercile (Low, Medium, and High), we report the average level of insider trading activity measured using NIP (Panel A) and OI (Panel B) across three event windows: 30 days before the dividend announcement date, between the announcement date and the ex-dividend date (dividend window), and 30 days after the ex-dividend date. For each tercile, the first row shows the mean value of the insider-trading measure, and the second row reports the corresponding t-statistic. We also report a High–Low spread and t-statistics testing whether these High–Low differences are statistically different from zero.

Panel A: NIP

Tercile of Returns	30 Days Before Ann. Date	Between Ann. Date and Ex-Date	30 Days After Ex-Date
1 (Low)	-0.45	-0.23	-0.12
	(-15.25)	(-7.12)	(-3.50)
2	-0.38	-0.39	-0.15
	(-12.14)	(-11.25)	(-4.55)
3 (High)	-0.25	-0.46	-0.29
	(-8.10)	(-16.99)	(-9.67)
High - Low	0.20	-0.23	-0.17
T-Statistic	(4.83)***	(-5.56)***	(-3.88)***

Panel B: OI

Tercile of Returns	30 Days Before Ann. Date	Between Ann. Date and Ex-Date	30 Days After Ex-Date
1 (Low)	0.28	0.39	0.45
	(18.91)	(24.81)	(27.31)
2	0.33	0.31	0.44
	(21.21)	(17.65)	(27.21)
3 (High)	0.39	0.28	0.37
	(25.58)	(20.77)	(24.56)
High - Low	0.11	-0.11	-0.08
T-Statistic	(5.17)***	(-5.42)***	(-3.76)***

Table 7: Dividend Window Returns and Insider Trading

This table presents results of insider trading on returns. Returns are calculated from one day after the dividend announcement date to one day before the ex-dividend date (Dividend Window). Two measures of insider trading are used for each time period: NIP and OI. These are measured 30 days before the announcement date (columns 1 and 2), between the announcement and ex-date (columns 3 and 4), and 30 days after the ex-date (columns 5 and 6). All variables are defined in the appendix. Statistical significance at the 1%, 5%, and 10% confidence levels are denoted by *, **, and ***, respectively.

Time Period	30 Days Before Ann. Date		Between Ann. and Ex-Date		30 Days After Ex-Date	
	NIP	OI	NIP	OI	NIP	OI
Returns	0.7053***	0.3738***	-1.1654***	-0.5727***	-1.4601***	-0.7222***
	(2.84)	(3.02)	(-5.58)	(-5.56)	(-6.00)	(-5.98)
Size	-0.2643***	-0.1271***	-0.2085***	-0.1021***	-0.2179***	-0.1045***
	(-14.59)	(-14.05)	(-10.89)	(-10.65)	(-11.76)	(-11.39)
Leverage	-0.0391	-0.0296	0.0911	0.0498	-0.1777	-0.0886
	(-0.31)	(-0.47)	(-0.76)	(-0.83)	(-1.47)	(-1.49)
Profitability	-3.3069***	-1.6395***	-3.7500***	-1.7536***	-2.8677***	-1.3675***
	(-4.49)	(-4.47)	(-4.81)	(-4.49)	(-3.86)	(-3.73)
Asset Growth	0.0547	0.0198	0.1964**	0.1055**	0.0369	0.0401
	(-0.64)	(-0.48)	(-2.13)	(-2.33)	(-0.45)	(-0.99)
Number of Years	-0.9356**	-0.4751**	0.2046	0.1199	-0.4481	-0.2729
	(-2.52)	(-2.55)	(-0.57)	(-0.69)	(-1.05)	(-1.25)
Observations	2,726	2,726	2,578	2,578	2,729	2,729
Adjusted R-Sq.	0.366	0.368	0.339	0.340	0.342	0.347
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Table 8: Dividend Window Returns by Size & Profitability

This table presents regression results of market-adjusted returns on dividend yield, interest rate, and past market returns. Panel A includes interaction terms between the explanatory variables and indicators for firms below the median in size and profitability, allowing for heterogeneity in effects by firm type. This approach helps distinguish between a rational preference for dividend-paying firms and behavior consistent with the free dividend fallacy. Returns are calculated from one day after the dividend announcement date to one day before the ex-dividend date (Dividend Window). Columns (1) to (4) vary by fixed effects and interaction specifications. White robust standard errors are used, and statistical significance at the 1%, 5%, and 10% levels is denoted by ***, **, and *, respectively. All variables are defined in the appendix.

Panel A: Returns by firm type: main and interaction effects

	(1)	(2)	(3)	(4)
Dividend Yield (DY)	0.0104***	0.0105***	0.0132***	0.0117***
	(4.09)	(3.71)	(5.31)	(4.59)
Interest Rate (IR)	-1.2110**	-5.8621***	-1.4453**	-5.9439***
	(-2.28)	(-2.81)	(-1.98)	(-2.67)
Market Return (MR)	-0.0170***	-0.0282***	0.0093	-0.0108
	(-2.77)	(-2.99)	(1.13)	(-0.97)
DY * Low Profitability	0.0038	0.0027		
	(1.06)	(0.71)		
IR * Low Profitability	-0.9869	-0.6422		
	(-1.39)	(-0.82)		
MR * Low Profitability	0.0225**	0.0086		
	(2.19)	(0.79)		
DY *Low Size			-0.0020	0.0007
			(-0.53)	(0.17)
IR * Low Size			-0.1356	-0.2738
			(-0.16)	(-0.26)
MR * Low Size			-0.0292***	-0.0251**
			(-2.78)	(-2.11)
Controls	Yes	Yes	Yes	Yes
Observations	10,940	10,940	10,940	10,940
Adjusted R-squared	0.0134	0.0456	0.0149	0.0469
Year FE	No	Yes	No	Yes
Firm FE	No	Yes	No	Yes

Table 8 (cont.): Dividend Window Returns by Size & Profitability

Panel B reports the combined coefficients of each explanatory variable (Dividend Yield, Interest Rate, and Market Return) and their respective interaction terms. For each, the joint F-statistic and associated p-value test the null hypothesis that the combined effect is equal to zero. All specifications correspond to those shown in Panel A.

Panel B: Joint significance of main variables and interactions				
	(1)	(2)	(3)	(4)
DY + DY Interaction	0.0142	0.0132	0.0112	0.0124
F-stat	27.0247	22.1572	14.5068	13.5004
P-value	0.0000	0.0025	0.0020	0.0003
IR + IR Interaction	-2.1979	-6.5043	-1.5809	-6.2177
F-stat	14.4010	9.1214	9.5436	8.9717
P-value	0.0001	0.0704	0.0026	0.0002
MR + MR Interaction	0.0055	-0.0196	-0.0199	-0.0359
F-stat	0.4312	3.2753	9.0758	13.3729
P-value	0.5114	0.0000	0.0001	0.0027

Appendix: Variable Definitions

This table describes each variable used throughout the analysis. All variables are winsorized at the 1st and 99th percentile. We restrict the sample to REITs that pay quarterly dividends. The window of time between the announcement date and ex-date for monthly dividend firms is small, and monthly dividend-paying firms would be oversampled in our regressions.

Variable	**Definition**
Returns	Cumulative returns between one day after the dividend announcement date and one day before the ex-dividend date.
Dividend Yield	Dividend yield of the REIT as of the end of the month prior to the dividend announcement date. Following Welch and Goyal, 2008 and Goyal et al., 2024, Dividend Yield (d/y) is the difference between the log of aggregate 12-month dividends and the log of lagged prices.
Interest Rate	1-month T-bill rate as of the end of the month prior to the dividend announcement date. Obtained from the economic research database of the Federal Reserve Bank of St. Louis.
Market Return	Aggregate return on the CRSP value-weighted index during the 12 months prior to the announcement date.
Size	Log of total assets of the firm as of the last fiscal year end.
Leverage	Total debt divided by total assets as of the last fiscal year end.
Profitability	Net income divided by total assets as of the last fiscal year end.
Asset Growth	Year-over-year asset growth as defined by Fama and French, 2015.
Number of Days	Number of days between announcement date and ex-date.
Market Adjusted Returns	Total return of the REIT minus the return of the S&P REIT Index for various intervals (3, 6, 9, and 12 months).
GICS Indicator	Indicator that takes the value of 1 after September 2016 and 0 otherwise.
Net Insider Purchase (NIP)	Aggregate number of shares bought minus shares sold, normalized by total insider trades during the window: $$NIP_{i,j,t} = \frac{\sum_{j=1}^{n} \#shares\,buy_{i,j,t} - \sum_{j=1}^{n} \#shares\,sell_{i,j,t}}{\sum_{j=1}^{n} \#shares\,buy_{i,j,t} + \sum_{j=1}^{n} \#shares\,sell_{i,j,t}}$$
Ownership Increase (OI)	Proportion of insider buyers during the window: $$OI_{i,t} = \frac{\sum_{i=1}^{n} \#insider\,buyer_{i,t}}{\sum_{i=1}^{n} \#insider\,buyer_{i,t} + \sum_{i=1}^{n} \#insider\,seller_{i,t}}$$